

# Metals and Alloys Worksheet 1

## Nichromes

### Introduction

Nichrome is the glowing heart of dryers and toasters. There is a high chance that you have used products that contain nichromes at home. Some everyday products function in an effective way because of the properties of nichromes.

### Aims

The aim of this project is to introduce you to some of the properties and uses of these alloys by exploring the information related to nichromes in the research literature and databases.

### Task

Using literature available in books, research papers and databases found online in the Library and more widely on the internet conduct a research on Nichromes alloys, its properties and processing by answering the following questions:

1. What are Nichromes? What is the chemical composition and microstructure of these alloys? Could you comment on the type of chemical bond that is present in these alloys?
2. When where Nichromes described? Could you provide a short summary of the historical development of Nichromes?
3. What are the main properties of Nichromes? What properties makes Nichromes a valuable material? What does service temperature mean and how is this important when discussing the applications of Nichromes?
4. Are Nichromes in the 'critical materials' list? What is the definition of a critical material?
5. Which compounds increase oxidation resistance to Nickel alloys? Perform a short search using databases and information on the internet.
6. Give at least 2 examples of the use of Nichromes in everyday products that do not include the examples in the description of this project. Could you comment on the economics and environmental impact of using Nichromes?

## **Report**

Prepare a 2-page report with your findings that contains the following sections:

1. Introduction
2. Historical development of Nichromes, chemical composition and methods of fabrication
3. Advantages and disadvantages of Nichromes alloys as materials, including use in everyday products
4. Summary of report
5. References

The questions presented in the task section can be used as a guide to find information that will be summarized in the sections above. You can include figures, diagrams and other graphical information as long as the size is large enough to be visible. Remember that diagrams and figures must be included in the text.

Please include all the references that you may have consulted for the development of the report.

The report should be prepared using Arial font, size 11. No other larger or smaller size can be used for the main text.

You can use Arial font size 10 for the references.

### **Assessment and deadline**

This report represents 10% of the total mark for this module. The report will be assessed on the technical and English merit. Check your spelling for any mistakes.

**The deadline is: 10/03/2020**

Please note: We will contact you if there are any problems with uploading coursework to QMPlus and provide alternative means of submission.

**DO NOT submit your coursework by email**, it will not be considered for marking.

Feedback and marks will be provided online by QMPlus